

AI-Assisted Architectural Study of a SKY130 130 nm Phase-Locked Loop: Block Analysis, Toolchain Setup, and Debugging Strategy

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Abstract—This report documents a structured, AI-assisted study of a 130 nm SKY130-process phase-locked loop (PLL) implemented in the open-source reference repository nitjsr_pll_130nm. The objective of Week 1 is conceptual mastery rather than full reproduction: understanding PLL sub-block architecture, feedback loop mathematics, toolchain requirements (xschem / ngspice / Magic / OpenLane), pre-layout versus post-layout simulation differences, and systematic debugging strategies. Every insight recorded here has been independently verified against the reference repository and primary literature.

I. INTRODUCTION

Phase-Locked Loops are essential mixed-signal circuits used for clock synthesis, frequency multiplication, and jitter reduction. The reference repository targets an $8\times$ clock multiplier on the SKY130 open-source PDK: a 5 MHz reference input is multiplied to produce a 40 MHz locked output [1]. The entire flow uses only free and open-source tools, making it an ideal learning platform.

This week's work decomposes the repository into independently understandable tasks, uses AI prompting to accelerate reading of netlist and simulation files, and produces a verified block-by-block understanding of what inputs, outputs, and failure modes to expect.

II. PLL ARCHITECTURE OVERVIEW

The $8\times$ PLL comprises five major sub-blocks connected in a negative-feedback loop. The reference clock $f_{ref} = 5$ MHz and the divided-down VCO output f_{div} are compared by the Phase Frequency Detector (PFD). The resulting UP/DOWN pulses drive the Charge Pump (CP), which sources or sinks current into the Loop Filter (LF). The resulting control voltage V_{ctrl} tunes the Voltage-Controlled Oscillator (VCO), and a $\div 8$ Frequency Divider closes the feedback path [1].

Flow: $f_{ref} \rightarrow \text{PFD} \rightarrow \text{CP} \rightarrow \text{LF} \rightarrow \text{VCO} \rightarrow f_{out}$, with $\div 8$ Divider feeding back to PFD.

The fundamental frequency relationship is:

$$f_{out} = N \cdot f_{ref} \rightarrow 40 \text{ MHz} = 8 \times 5 \text{ MHz}$$

III. SUB-BLOCK ANALYSIS

A. Phase Frequency Detector

The PFD is a digital state machine, typically built from two D-flip-flops with an AND-gate reset path. When f_{ref} leads f_{div} , it asserts UP; when f_{div} leads, it asserts DOWN. A key failure mode is the dead zone: if reset propagation is too fast, narrow pulses near lock are suppressed, causing phase noise. In xschem schematics, the PFD is often drawn as a custom cell referencing SKY130 DFF primitives.

B. Charge Pump

The CP converts digital UP/DOWN pulses into an analog current. An ideal CP sources I_{CP} on UP and sinks the same on DOWN. Mismatch between source and sink currents is a primary cause of reference spurs. In the SKY130 netlist, PMOS transistors act as current sources and NMOS as current sinks, biased by cascode mirrors to improve output impedance.

C. Loop Filter

A passive second-order RC filter (R_1, C_1, C_2) converts CP current pulses to V_{ctrl} . The zero introduced by R_1C_1 stabilises the loop. The loop bandwidth is approximately:

$$\omega_n = \sqrt{(K_{VCO} \cdot I_{CP}) / (2\pi \cdot N \cdot C_1)}$$

Stability requires phase margin $> 45^\circ$. Choosing $C_2 \approx C_1/10$ keeps the second pole well above ω_n .

D. Voltage-Controlled Oscillator

The VCO is a current-starved ring oscillator with an odd number of inverter stages. V_{ctrl} sets the tail-current bias, linearly shifting oscillation frequency. The VCO gain K_{VCO} (MHz/V) is extracted from a V_{ctrl} DC sweep. Ring oscillators on SKY130 exhibit significant supply-noise sensitivity; a replica-bias stage reduces this.

E. Divide-by-8 Frequency Divider

Three cascaded toggle flip-flops implement $\div 2, \div 4, \div 8$ division. The output of the third stage feeds f_{div} back to the PFD. Layout placement of the

divider close to the PFD minimises routing skew, which is critical for lock accuracy.

IV. SKY130 PDK & TOOLCHAIN SETUP

Tool	Role
xschem	Schematic capture
ngspice	SPICE simulation
Magic VLSI	Layout & DRC
netgen	LVS check
OpenLane	RTL→GDS flow

PDK setup requires setting the environment variable PDK_ROOT to the SKY130 installation path. The most common setup failure is a version mismatch between the PDK libraries and the installed tools — resolved by pinning versions via the volare PDK manager [2].

V. PRE-LAYOUT VS. POST-LAYOUT SIMULATION

Pre-layout simulations use ideal wire connections from the xschem netlist and confirm circuit functionality. Post-layout simulations use an RC-extracted netlist generated by Magic's ext2spice tool. Parasitic capacitances on VCO nodes directly degrade oscillation frequency and worsen phase noise. Expected differences: (a) VCO free-running frequency shifts by 5–20%; (b) lock time increases; (c) charge pump glitches may appear due to gate capacitance on switching nodes.

VI. SIMULATION WORKFLOW & DEBUG STRATEGY

Recommended incremental simulation strategy: (1) Verify VCO free-running range with a V_{ctrl} DC sweep before any loop closure. (2) Test PFD UP/DOWN pulse widths with a two-clock transient stimulus. (3) Verify CP current balance by probing $I(V_{ctrl})$ node directly. (4) Close the loop and run a full transient to observe lock. (5) For post-layout, compare V_{ctrl} settling waveform with pre-layout to quantify parasitic impact.

Common failure modes and fixes:

Symptom	Likely Cause
VCO not oscillating	Bias current too low
PLL never locks	fref outside VCO range
SPICE convergence error	Parasitic short in netlist
LVS mismatch	Floating port in layout

VII. CONCLUSION

This structured decomposition of the nitjsr_pll_130nm repository provides a clear pathway from PLL theory to open-source implementation. Each sub-block has defined inputs, outputs, and testable failure modes. The AI-assisted workflow accelerated literature cross-referencing and netlist reading, while human verification ensured correctness. Weeks 2–4 will proceed with incremental simulation, layout, and full post-extraction verification.

REFERENCES

- [1] H. Singh, "nitjsr_pll_130nm: SKY130 $8\times$ Clock Multiplier PLL," GitHub, 2023. [Online]. Available: https://github.com/himansh107/nitjsr_pll_130nm
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- [3] R. Best, Phase-Locked Loops: Design, Simulation, and Applications, 6th ed. McGraw-Hill, 2007.
- [4] B. Razavi, Design of Analog CMOS Integrated Circuits, 2nd ed. McGraw-Hill, 2017.